CONSTRUCTION STANDARDS: THAI VS WESTERN

Approved project standard (Oct 11 2026; walls revised Oct 12 2026). Applies to every sheet and to the element properties in the 3D model.

TOPIC	COMMON THAI PRACTICE	OUR STANDARD	WHY
Structure	RC frame with brick infill; timber roof/floor framing common	RC frame (columns, beams, slabs); steel roof framing; no timber framing	Termites and humidity destroy timber; RC + steel is durable and insurable
Exterior Walls	Single-skin red clay brick or thin block, cement render, no insulation	LGS stud infill in RC frame, mineral-wool cavity, fiber-cement sheathing, ventilated rainscreen (white lime render or charred timber)	Rainscreen + insulation cut heat gain and AC load; termite-proof, light, straight
Interior Partitions	Clay brick / block, rendered	LGS studs with moisture-resistant fiber-cement board (SCG Smartboard / Shera); double stud + 2 layers board at studio + bedrooms	Lighter, straighter, quieter; termite-proof; easy service routing
Wall Finish	Cement render, emulsion paint	AR mesh tape, lime plaster skim, textured finish (TBD), mineral paint warm white	Crack-resistant joints; breathable, resists mould
Concrete Walls	Poured where convenient	RC only for frame, pool, retaining walls, cisterns, engineer-required shear walls	Less embodied carbon and heat mass where not needed
Roof	Metal sheet or tile on timber/steel, foil at best, little insulation	Metal over radiant barrier, 75 mm insulation, ventilated cavity	Hot-roof heat is the #1 cooling load; quiet in monsoon rain
Traps	Bottle traps or none; floor drains often untrapped (smell)	P-trap on every fixture; deep-seal trapped floor drains with trap primers	Stops sewer gas, insects and odours
Venting	Often unvented; drains glug and siphon traps	Proper vent stack system through the roof	Keeps trap seals; fast, quiet drainage
Grease	Kitchen waste straight to sewer/septic	Kitchen grease interceptor GT-1 before septic	Protects septic + ATU and drain field
Water Supply	PVC cold + mixed fittings, rarely tested	PPR fusion-welded, pressure-tested before walls close	No hidden leaks; rated for hot water
Electrical	Partial earthing, RCD on some circuits, ad-hoc panel	TN-C-S earthing, 30 mA RCBO on all circuits, full panel schedule	Shock + fire safety in a wet climate
Wet Areas	Tile on render; leaks common	Waterproofing membrane in all wet areas, flood-tested 24 h	Prevents leaks and mould into stud cavities
Hot Water	Point-of-use electric heaters at each shower	Central heat-pump water heater + recirculation	~3x more efficient; one serviceable unit; instant hot water

WALL TYPES

- W1 Steel Stud Exterior: 150 mm LGS infill in RC frame, mineral wool, fiber-cement sheathing, ventilated rainscreen (white lime render or charred timber); inside fiber-cement + lime plaster, painted
- W2 Steel Stud: 92 mm LGS @ 400 mm, fiber-cement board both sides, mineral wool, lime plaster, painted; cement board + membrane on wet faces
- W3 Acoustic Stud: Double LGS / resilient channel, 75 mm mineral wool, 2 layers fiber-cement each side (studio + bedrooms)
- W4 RC Wall: Poured RC: pool, retaining walls, cistern, engineer-required shear walls only

WEBERMAN RESIDENCE

Mountain courtyard house, Thailand
Owner: Max Weberman

ISSUE

Design Development

NOT FOR CONSTRUCTION

DATE
11 Oct 2026

SCALE
NTS

MODEL
bim.json (from plan.json)

UNITS
Feet + inches (mm for members)

REVISION
DD-01

NOTES

Western standard = trapped + vented DWV, tested PPR, full earthing + RCBO, membranes, central hot water.

Engineer to confirm shear walls (W4) and all member sizes.

SHEET TITLE

Construction Standards (Thai vs Western)

SHEET

G-003